

Vapor pressures in the ternary system water-nitric acid-sulfuric acid at low temperature: A reexamination

Djamel-Eddine Taleb, Jean-Luc Ponche, and Philippe Mirabel

Centre de Géochimie de la Surface/Centre National de la Recherche Scientifique and Université Louis Pasteur, Strasbourg, France

Abstract. The equilibrium partial vapor pressures over the binary solutions $\text{H}_2\text{O}/\text{HNO}_3$, $\text{H}_2\text{SO}_4/\text{H}_2\text{O}$, $\text{HNO}_3/\text{H}_2\text{SO}_4$ and the ternary mixture $\text{H}_2\text{SO}_4/\text{HNO}_3/\text{H}_2\text{O}$ have been re-determined for temperatures below 25°C . The model presented here is valid over the entire range of concentrations, for temperatures in the range 190–298 K. The predictions of this model are compared with the existing measurements and other predictions, both at room temperature and at low temperatures down to about 190 K. The calculated partial pressures agree very well with the available experimental data

1. Introduction

The knowledge of accurate partial vapor pressures and total vapor pressures over aqueous solutions is very important in modeling environmental processes. They are needed to predict solubilities of gases in liquids, in computing aerosol rate of nucleation and growth, etc. Very recent interest has developed also in the formation of the solid phase in the higher atmosphere, and the accurate prediction of the phase transitions also requires the knowledge of thermodynamic data including, again, the equilibrium partial pressures of the constituents. Among the various aqueous solutions which are of some concern in describing atmospheric processes, mixtures of water with nitric acid and/or sulfuric acid play a major role. The water-sulfuric acid system is frequently involved in the formation of small acidic aerosols through heteromolecular nucleation, while the importance of the water-nitric acid mixture is evident in the formation of polar stratospheric clouds.

Until very recently, most of the experimental determinations of the partial pressures (or activities or activity coefficients) were performed over a limited range of concentrations and at temperatures much higher than those prevailing in the upper atmosphere. The data then had to be extrapolated over a wide range of concentrations and temperatures, leading to large had uncertainties.

One way of predicting thermodynamic properties in multicomponent mixtures is to use some existing rules or models such as the Van Laar theory, the regular solution theory, the Debye-Hückel theory, etc. [e.g., *Prausnitz*, 1969]. Care must also be taken to verify that the predicted values are self-consistent and satisfy the Gibbs-Duhem or Gibbs-Margules equations.

Several predictions of the partial pressures at low temperatures in the binary $\text{H}_2\text{SO}_4\text{-H}_2\text{O}$ and $\text{HNO}_3\text{-H}_2\text{O}$ mixtures have been carried out in the past. For example, we have previously investigated the $\text{HNO}_3\text{-H}_2\text{O}$ system for temperatures in the range -10 to 70°C [*Clavelin and Mirabel*, 1979]. Although we have later

slightly modified these predictions [*Jaecker-Voirol et al.*, 1990], the extrapolation of these earlier values to lower temperatures was used by *Crutzen and Arnold* [1986] and by *Toon et al.* [1986] to predict the formation of polar stratospheric clouds.

We also investigated the equilibrium vapor pressures for the $\text{H}_2\text{SO}_4\text{-H}_2\text{O}$ mixture, [*Jaecker-Voirol et al.*, 1990] but since our work was published, two other compilations appeared [*Zelevnik*, 1991; *Bolsaitis and Elliott*, 1991], in addition to some experimental determinations performed at low temperature [*Zhang et al.*, 1993a].

Finally, although the ternary mixture $\text{HNO}_3\text{-H}_2\text{SO}_4\text{-H}_2\text{O}$ is very important in the atmosphere, the predictions concerning its thermodynamic properties at low temperatures are rather scarce. We have already published a work on this subject [*Jaecker-Voirol et al.*, 1990], but since it appeared in the literature, new experimental results concerning the equilibrium partial pressures of HNO_3 and H_2O for this mixture became available [*Zhang et al.*, 1993b].

The aim of this study is therefore to produce new estimations of the thermodynamic properties (i.e., partial vapor pressures, activities, and activity coefficients) of the binary systems $\text{HNO}_3\text{-H}_2\text{O}$ and $\text{H}_2\text{SO}_4\text{-H}_2\text{O}$ and of the ternary system $\text{H}_2\text{O-HNO}_3\text{-H}_2\text{SO}_4$, based on the knowledge of the corresponding properties in the binary mixtures. In addition, the data are presented in a very simple form, allowing very easy calculations of the different equilibrium partial pressures, both for the binary mixtures and for the ternary system, over the entire range of concentrations, and for the temperatures of interest in the atmosphere.

2. The $\text{H}_2\text{O-HNO}_3$ System

2.1. Results

The results for this system (hereafter referred to as system I) at low temperatures, have already been presented in Table 3 of the article by *Jaecker-Voirol et al.* [1990]. The results were also compared favorably in Figures 1 and 2 of the same article with the experimental determinations by *Hanson and Mauersberger* [1988] for two mole fractions of HNO_3 , namely, 0.4 and 0.25.

In the present work, no modifications are made to the previous results of *Jaecker-Voirol et al.* [1990] for the $\text{H}_2\text{O-HNO}_3$ system. However, in order to predict more easily the partial vapor

pressures in the ternary system H₂O-HNO₃-H₂SO₄, we shall present the data for this binary system in terms of a Van Laar type equation, without any appreciable loss of precision. According to this type of equation, the activity coefficients γ_1 and γ_2 of components 1 and 2 in a binary system can be expressed as (derivation of this equation is given in the appendix):

$$T \log \gamma_{1,1} = \frac{A_{1,1} X_2^2}{[X_2 + B_{1,1} X_1]^2} \quad (1a)$$

$$T \log \gamma_{1,2} = \frac{A_{1,2} X_1^2}{[X_1 + B_{1,2} X_2]^2} \quad (1b)$$

where T is the temperature in degrees K, X the mole fraction, and $A_{i,j}$ and $B_{i,j}$ are constants for a given system (note that they may vary with temperature). The derivation of the Van Laar equations suggests that such equations should be used for solutions of relatively simple, nonpolar liquids. However, it has been found empirically that these equations are frequently able to represent activity coefficients of more complex solutions, such as our mixtures, if one regards A and B as adjustable parameters [Prausnitz, 1969].

From the general definitions of the activity coefficient γ_i and of the activity a_i :

$$a_i = \frac{P_i}{P_{0,i}} \quad (2a)$$

$$\gamma_i = \frac{a_i}{X_i} = \frac{P_i}{X_i P_{0,i}} \quad (2b)$$

it becomes easy to compute the partial pressure P_i of i , provided that the equilibrium pressure of pure i , $P_{0,i}$, is known, which is generally the case

$$P_i = \gamma_i X_i P_{0,i} \quad (3)$$

The equilibrium vapor pressure of pure nitric acid, which was taken from *Duisman and Stern* [1969] is given by

$$\log P_{0,\text{HNO}_3} (\text{torr}) = 7.61628 - \frac{1486.238}{(T - 43)}, \quad (4)$$

while that of water, expressed in millibars for the temperature range 220 to 320 K, is taken from the compilation of *Tabata* [1973]:

$$\log P_{0,\text{H}_2\text{O}} (\text{mbars}) = 8.42926609 - \frac{1827.17843}{T} - \frac{71208271}{T^2} \quad (5)$$

Reworking the data generated by *Clavelin and Mirabel* [1979], *Jaeger-Voirol et al.* [1990], one finds the following values for the constants $A_{1,1}$ and $B_{1,1}$ for water (component 1), $A_{1,2}$ and $B_{1,2}$ for HNO₃ (component 2) to be incorporated in equations (1a) and (1b):

$$\begin{aligned} A_{1,1} &= -391.43 - 7.44 \times 10^4 / T \\ A_{1,2} &= -627.739 - 1.406 \times 10^5 / T \\ B_{1,1} &= 1/B_{1,2} = 0.5695. \end{aligned} \quad (6)$$

2.2. Comparison With Other Results

We present in Figure 1, the comparison of our calculations at 20°C with those deduced from the recent compilation of *Clegg and Brimblecombe* [1990], which covers the entire range of concentration. Also included in this figure are the experimental values of *Vandoni and Laudy* [1952] for the same temperature. These latter authors covered a large range of concentration, and tested the self-consistency of their measurement by use of the Duhem-Margules equation. The comparison of the three sets of data is indeed very good. The same conclusion can be drawn when comparing at 25°C (Figure 1) the present compilation, *Clavelin and Mirabel* [1979] results, *Clegg and Brimblecombe* [1990] predictions, and the compilation of *Redlich et al.* [1968]. It is then difficult to understand why *Clegg and Brimblecombe* wrote in their article that "the two previous analysis (particularly that of *Clavelin and Mirabel*) appear to be in error at high concentrations."

When *Clavelin and Mirabel* [1979] and later *Jaeger-Voirol et al.* [1990] published their results, there were few data available for the H₂O-HNO₃ system at temperatures below 20°C. That situation has changed now, since both new experimental data and theoretical predictions are available. The experimental determinations of the equilibrium partial pressures of H₂O and HNO₃ are those of *Hanson and Mauersberger* [1988], while the theoretical predictions are provided by *Brimblecombe and Clegg* [1990].

Hanson and Mauersberger measured the vapor pressure of water and of the acid for a large range of temperatures (depending upon the concentration), and for seven acid concentrations, ranging from $X_{\text{HNO}_3} = 0.1$ to 0.47. As mentioned previously, *Hanson and Mauersberger's* data, at least for $X_{\text{HNO}_3} = 0.25$ and 0.4 compare very favorably with the compilation of *Jaeger-Voirol et al.* [1990] over a wide range of temperatures. In Figure 2, we compare the partial pressures of H₂O and HNO₃ deduced from the predictions of the present compilation with the measurements of *Hanson and Mauersberger* for the seven acid concentrations. As can be seen from this figure, the agreement between the calculated partial pressures and the experimental data is always very good, with a slight exception for the partial pressure of the acid when $X_{\text{HNO}_3} = 0.1$. In that case, the experimental data lie above the predicted values, and the slope is somewhat different. We note, however, that the experimental slope for this case is based on only two data points and is therefore subject to large uncertainties.

In the same figure, we compare our predictions to those of *Brimblecombe and Clegg* [1990]. These authors, using another study [*Clegg and Brimblecombe*, 1990] based on a model developed by *Pitzer and Simonson* [1986] were able to provide equilibrium partial pressures for X_{HNO_3} ranging from 0 to 1 and for temperatures as low as - 60°C. As can be seen in Figure 1, from the comparison of the two sets of curves, the agreement between the two predictions appears to be very good.

To get a deeper insight at this comparison, we have calculated the maximum deviation of the two predictions, compared to the experimental data of *Hanson and Mauersberger*, for the seven concentrations and for both the water and acid partial pressure. Of course, this comparison is only valid within the temperature range of the experimental data. One first notes that, in general, the maximum deviations occur for the lower temperatures. In addition, for the acid, the maximum deviation occurs at 190 K for $X_{\text{HNO}_3} = 0.1$ between the experimental data and the prediction of *Brimblecombe and Clegg* (59% while the present calculation differs by 14% at most). For all the other concentrations, whatever

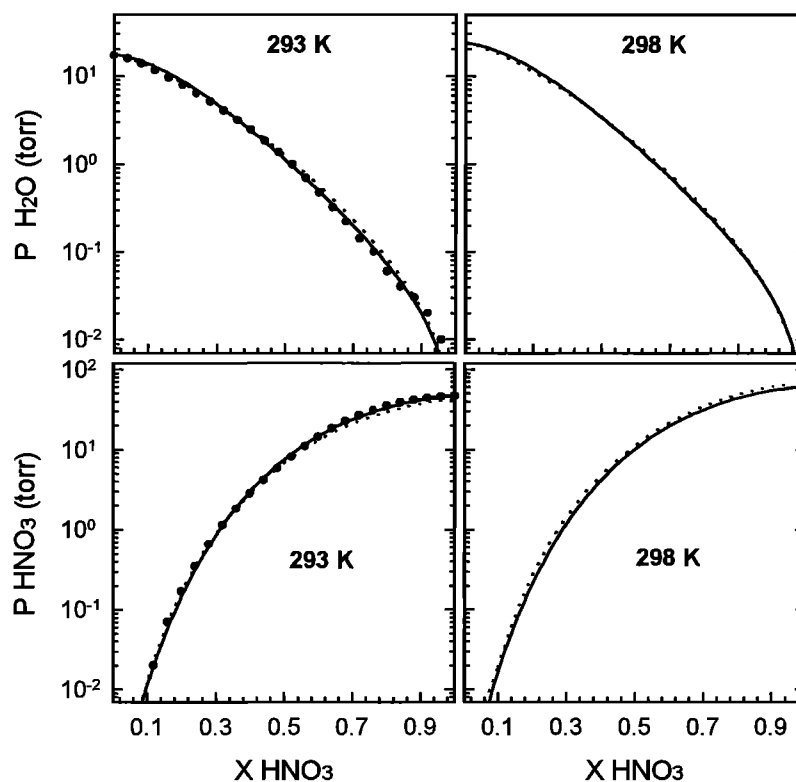


Figure 1. Comparison of P_{H_2O} and P_{HNO_3} (left panel, 293 K; right panel, 298 K) as a function of the mole fractions of HNO₃. The solid lines are the present work, dotted lines are Brimblecombe and Clegg [1990] predictions, and symbols represent Vandoni and Laudy [1952] experimental results. The data of Clavelin and Mirabel [1979] and of Redlich et al. [1968] at 298 K are not depicted, since they fall exactly on the curves of the present work.

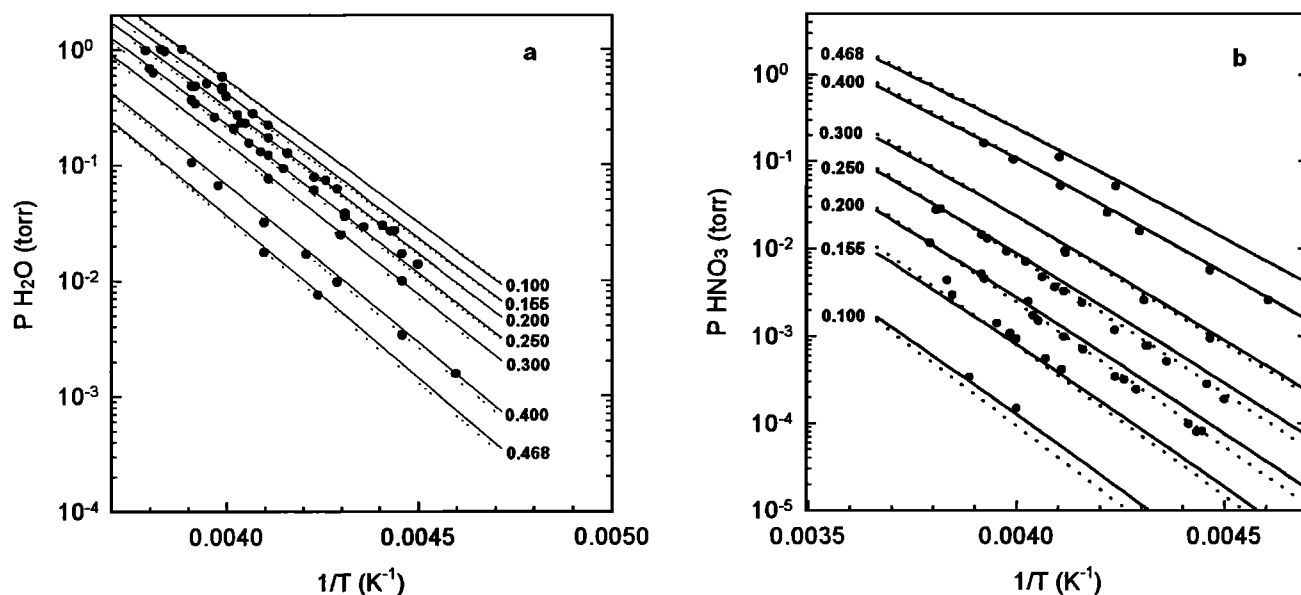


Figure 2. Vapor pressure of (a) H₂O and (b) HNO₃ for liquid and supercooled H₂O/HNO₃ mixtures. Mole fractions of HNO₃ are indicated in the figures. The solid lines are the present work, dotted lines are Brimblecombe and Clegg [1990] predictions, and symbols represent Hanson and Mauersberger [1988] experimental results.

is the prediction, the maximum deviations range from a few percent for the highest temperatures to about 30% at most for the lowest. For the water partial pressure, the maximum deviations, i.e., 20%, occur between the experimental measurements and the predictions of Brimblecombe and Clegg, for $X_{\text{HNO}_3} = 0.25$ and 0.468. Concerning our predictions, the maximum deviation reaches only 12% for $X_{\text{HNO}_3} = 0.25$ and otherwise remains below a few percent.

In conclusion, it is remarkable to see that our predictions, based on an extremely simple model expressed by only one equation and two parameters, is able to predict accurately the variation of the vapor partial pressures of each constituent for a wide range of temperatures and concentrations.

3. The H₂O-H₂SO₄ System

3.1. Results

We already published, a few years ago, the equilibrium partial pressures of H₂O and H₂SO₄ at low temperatures [Jaeger-Voirol *et al.*, 1990]. These predictions were based on the thermodynamic compilation, derived at room temperature, by Giauque *et al.* [1960]. Partial pressures at low temperatures were then calculated by use of the Clausius-Clapeyron equation, assuming a constant heat capacity difference. Since we published our predictions, two other compilations appeared concerning this system [Zeleznik, 1991; Bolsaitis and Elliott, 1990], in addition to direct experimental determinations performed at low temperature [Zhang *et al.*, 1993a]. These authors compared their experimental measurements, for the partial pressure of water, to our former predictions for four concentrations ranging from 33.9 to 62.6 wt % of H₂SO₄ (their Figure 2), and the agreement was described as "poor."

We present in this article new predictions for the H₂O-H₂SO₄ (hereafter referred to as system II) based on the compilations of Giauque *et al.* [1960] and on the more recent work by Zeleznik [1991] and Bolsaitis and Elliott [1990]. It is interesting to note that Zeleznik as well as Bolsaitis and Elliott, although they criticize the work of Giauque *et al.*, present "corrected" data in disagreement with each other. For example, at 25°C, they both find relative chemical potentials of the acid in good agreement with that of Giauque *et al.*, while they both find disagreement for the relative chemical potentials of water. For example, although Zeleznik finds the maximum disagreement (7.5%) for a mole fraction $X_{\text{H}_2\text{SO}_4}$ of about 0.65, and good agreement for $X_{\text{H}_2\text{SO}_4}$ larger than 0.8, Bolsaitis and Elliott find a very good agreement for mole fractions lower than 0.8 while this agreement deteriorates for larger mole fractions, reaching a maximum deviation of 12.8% for $X_{\text{H}_2\text{SO}_4} = 0.89$.

In this new compilation, we chose to use the data of Zeleznik [1991], since this is the only data set available for low temperatures. Recalculating his data by using the pure liquids as references for the activities, and fitting them with a Van Laar type equation, the following values for the constant $A_{\text{II},1}$ and $B_{\text{II},1}$ for water (component 1), $A_{\text{II},3}$ and $B_{\text{II},3}$ for sulfuric acid (component 3) to be incorporated in equations (1a) and (1b) are

$$\begin{aligned} A_{\text{II},1} &= 2.989 \times 10^3 - 2.147 \times 10^6 / T + 2.33 \times 10^8 / T^2 \\ A_{\text{II},3} &= 5.672 \times 10^3 - 4.074 \times 10^6 / T + 4.421 \times 10^8 / T^2 \\ B_{\text{II},1} &= 1/B_{\text{II},3} = 0.527. \end{aligned} \quad (7)$$

The partial vapor pressures of sulfuric acid over the solution can then be calculated by using the vapor pressure of the pure acid given by Ayers *et al.* [1980]:

$$\ln P_{0,\text{H}_2\text{SO}_4} (\text{atm}) = -10156/T + 16.259 \quad (8)$$

3.2. Comparison With Other Results

In Figure 3, we compare the results of our calculation to those of Bolsaitis and Elliott [1990], Giauque *et al.* [1960], and Zeleznik [1991] for the activity of water, at a temperature of 298 K. As can be seen from this figure and according to the above discussion, the activities are generally in good agreement for mole fractions $X_{\text{H}_2\text{SO}_4}$ in the range 0 to 0.5, while this agreement deteriorates for larger $X_{\text{H}_2\text{SO}_4}$.

Recently, water vapor pressure measurements at low temperatures over liquid and supercooled liquid have been reported by Zhang *et al.* [1993a] for H₂SO₄ concentrations ranging from 20 to 70 wt %. We present in Figure 4 a comparison between our predictions and their results (plotted using the coefficients listed in their table I), for the eight concentrations they investigated. We note at first a generally good agreement between the two sets of curves, especially for the slopes. The maximum deviation is 26% for 44.2 wt %, and remains below 7% for the other concentrations. However, some differences appear for the concentrations around 35% and 70%. It seems that small inconsistencies, probably due to experimental uncertainties, exist for the data of Zhang *et al.* For example, at 250 K, they find a partial pressure of water of 4.77×10^{-1} torr for a solution containing 33.85 wt % of acid and 4.9×10^{-1} torr for a solution of 35.7 wt %, while in the second case the partial pressure of water should be lower since the concentration of acid is increased. The same behavior is found for the concentrations 67.8 and 70 wt %.

Zhang *et al.* [1993a] also provided an expression which fits their experimental measurements as a function of H₂SO₄ concentration (their Table II). The comparison between our predictions and their fitted values is given in Figure 5, for 250 and 200 K. As shown in this figure, our predictions are in good agreement with the H₂O vapor pressures of Zhang *et al.* for both temperatures. Very recently, Luo *et al.* [1995] and Clegg and

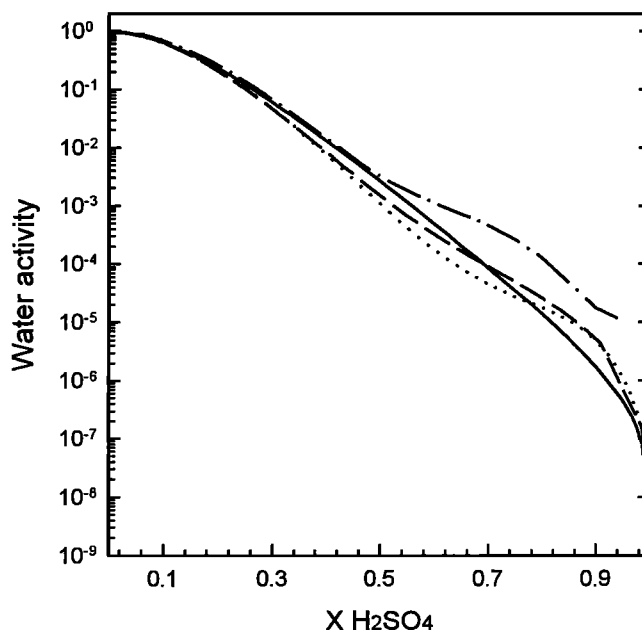


Figure 3. Activity of water in the H₂O/H₂SO₄ solution at 298 K. Solid line, our model; dashed line, Giauque *et al.*, [1960]; dash-dotted line, Bolsaitis and Elliott [1990]; dotted line, Zeleznik [1991].

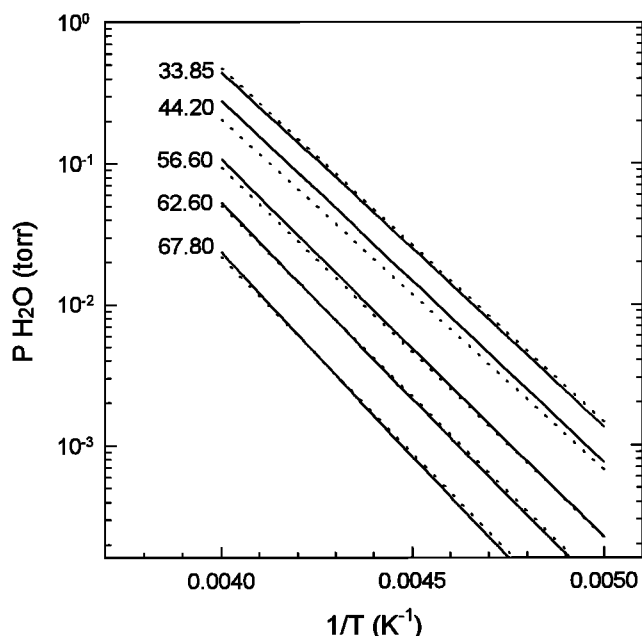


Figure 4. Vapor pressure of H₂O for liquid and supercooled H₂O/H₂SO₄ mixtures. Weight percentages of H₂SO₄ are indicated in the figure. The solid lines are the present work; dotted lines are taken from Zhang *et al.* [1993a].

Brimblecombe [1995] also compared the results of their model-predicted vapor pressures with the measurements of Zhang *et al.* [1993a]. In both cases, they found up to 45% deviations, and Clegg and Brimblecombe concluded that "it appears unlikely that all of this difference is attributable to inaccuracies in the models." In view of the general agreement between our predictions and Zhang *et al.*'s results, we cannot support this conclusion.

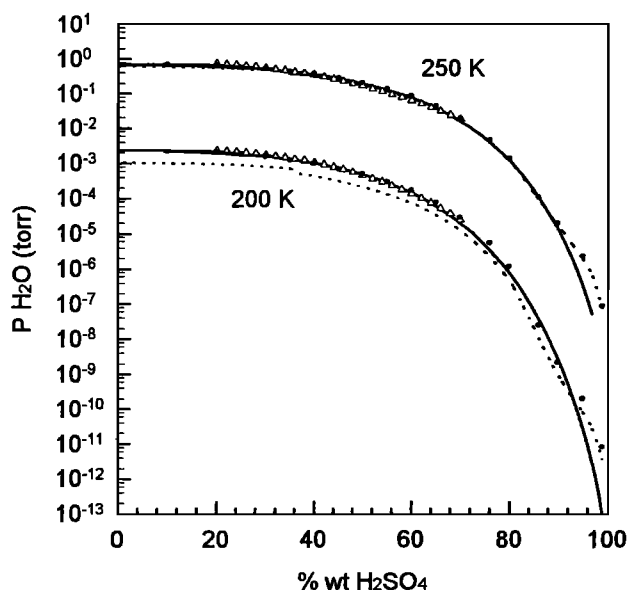


Figure 5. Vapor pressure of H₂O as a function of H₂SO₄ wt % at 200 and 250 K. The solid lines are the present work, the dashed lines are taken from Jaeger-Voirol *et al.* [1990], the triangles are Zhang *et al.* [1993a] experimental data, and the circles correspond to the predictions of Gmitro and Vermeulen [1964].

While in good agreement, at room temperature, with the predictions we already published [Jaeger-Voirol *et al.*, 1990], our present calculations differ significantly at low temperatures from these earlier results (see Figure 5). A reexamination of these data shows that the major differences may arise from the treatment of the variations of the heat capacity of the solution with temperature. This last observation was already pointed out to us by B. Luo and T. Peter (private communication, 1992). Also shown in Figure 5 are the widely used predictions of Gmitro and Vermeulen [1964]. As can be seen in this figure, these predictions for P_{H_2O} in the H₂O-H₂SO₄ system were very accurate, with the exception of the values corresponding to large concentrations in sulfuric acid.

We can draw again the same conclusion as for the H₂O-HNO₃ system; i.e., it is remarkable that our predictions, based on an extremely simple model expressed by only one equation and two parameters, is able to predict accurately the variation of the vapor partial pressures of each constituent for a wide range of temperatures and concentrations.

4. The HNO₃ - H₂SO₄ System

In order to predict the different partial vapor pressures in the ternary mixture HNO₃-H₂SO₄-H₂O, some information is needed on the HNO₃-H₂SO₄ system (hereafter referred to as system III). Unfortunately, only the vapor pressure of the nitric acid has been measured experimentally [Vandoni, 1944], and these measurements were limited to the single temperature of 273 K. Nevertheless, this allows the determination of the constants $A_{III,2}$ and $B_{III,2}$ for nitric acid and, consequently, of $A_{III,3}$ and $B_{III,3}$ for sulfuric acid in this mixture, but only for the temperature of 273 K.

The following constants are then obtained:

$$\begin{aligned} A_{III,2} &= -250.52 \\ A_{III,3} &= -100.21 \\ B_{III,2} &= 1/B_{III,3} = 0.4. \end{aligned} \quad (9)$$

There are no data available allowing us to determine the (weak) variation of the activity coefficients with temperature. In such a case, one of two simplifying approximations is usually made [Prausnitz, 1969]. The first is to assume that $\log \gamma$ is invariant with temperature, the second is to assume that $T \log \gamma = \text{const}$. According to Prausnitz, the second assumption generally provides a better approximation. Using this last approximation, it is then possible to generate the partial pressures of both H₂SO₄ and HNO₃ in the HNO₃-H₂SO₄ mixture, over the full range of concentrations.

Due to the rather large uncertainties in Vandoni's measurements, and to the assumption made above, the activity coefficients calculated from (1a)-(1b), and then the vapor pressures should be considered only as estimates.

5. The HNO₃-H₂SO₄-H₂O System

According to the relationships developed in the appendix and to the formula given below, it is possible to determine the activity coefficients and then the partial pressures of each of the constituents of the ternary mixture, over the entire range of concentrations, from the coefficients determined for the three binary systems:

$$T \log \gamma_{I,1} = \frac{A_{I,1}(X_2^2 + B_{III,2}X_2X_3) - \frac{1}{2}A_{III,2}B_{I,1}X_2X_3}{(B_{I,1}X_1 + X_2 + B_{III,2}X_3)^2} + \frac{A_{II,1}(X_3^2 + B_{III,2}X_2X_3) - \frac{1}{2}A_{III,3}B_{II,1}X_2X_3}{(B_{II,1}X_1 + B_{III,2}X_2 + X_3)^2} \quad (10a)$$

$$T \log \gamma_{I,2} = \frac{A_{I,2}(X_1^2 + B_{II,3}X_1X_3) - \frac{1}{2}A_{II,3}B_{I,2}X_1X_3}{(X_1 + X_2B_{I,2} + B_{II,3}X_3)^2} + \frac{A_{III,3}(X_3^2 + B_{II,1}X_1X_3) - \frac{1}{2}A_{II,1}B_{III,2}X_1X_3}{(B_{II,1}X_1 + B_{III,2}X_2 + X_3)^2} \quad (10b)$$

$$T \log \gamma_{I,3} = \frac{A_{II,3}(X_1^2 + B_{I,2}X_1X_2) - \frac{1}{2}A_{I,2}B_{II,3}X_1X_2}{(X_1 + X_2B_{I,2} + B_{II,3}X_3)^2} + \frac{A_{III,2}(X_2^2 + B_{I,1}X_1X_2) - \frac{1}{2}A_{I,1}B_{III,3}X_1X_2}{(B_{I,1}X_1 + X_2 + B_{III,3}X_3)^2} \quad (10c)$$

To test the validity of our calculations, we have compared them, first with the data of *Vandoni* [1944] at 273 K, and then with the data of *Zhang et al.* [1993b] at temperatures below 240 K.

Vandoni measured the H₂O and HNO₃ vapor pressures for 43 different compositions ranging from 5 to 90 wt % for HNO₃. In his article, Vandoni did not discuss the precision of his experiments, but from a latter publication [*Vandoni and Laudy*, 1952], one can learn that the uncertainties on this kind of measurement are of the order of ± 0.3 torr. The data of Vandoni clearly show the presence of a maximum for P_{HNO_3} , when the quantities of water and of sulfuric acid are the same, in terms of mole fractions. This fact was already observed at the beginning of this century by *Saposchnikow* [1904].

In Figures 6 and 7, we compare the results of the present calculations to the experimental values of Vandoni and to the recent compilation by *Carslaw et al.* [1995]. As can be seen from these figures, and taking into account the uncertainties on the experimental determinations, our calculated values are in very good agreements with those of Vandoni, for both $P_{\text{H}_2\text{O}}$ and P_{HNO_3} , over the full concentration range. In particular, our very simple model is able to predict the existence of the maximum of P_{HNO_3} , although the position of the maximum does not always correspond to the expected values, i.e., $X_{\text{H}_2\text{SO}_4}/X_{\text{H}_2\text{O}} = 1$. In addition, the calculated partial pressures in the ternary system tend smoothly to the values of the corresponding binary mixtures, when one of the component concentrations is reduced to zero.

The compilation by *Carslaw et al.* [1995], on the contrary, does not predict any maximum for nitric acid concentrations larger than 10% (Figure 7). These authors, in their Figure 6, also compared their predictions to Vandoni's data, and claimed that their partial pressures are in agreement within $\pm 40\%$ (with a mean absolute deviation of 17%). However, they restricted their comparison only to partial pressures smaller than 1.14 torrs. In

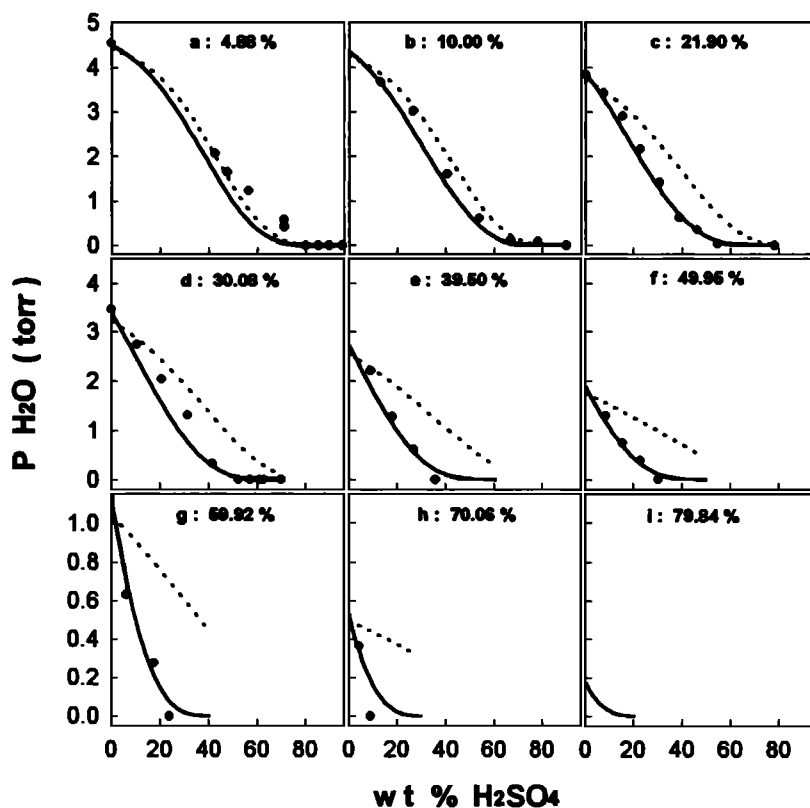


Figure 6. Comparison of the measured and predicted vapor pressures of H₂O over H₂O/HNO₃/H₂SO₄ mixtures at 273 K, plotted against wt % H₂SO₄. Weight percentages of HNO₃ are labeled in the figure. The symbols are *Vandoni*'s [1944] experimental data, solid lines are our fitted values, and dotted lines are *Carslaw et al.* [1995] predictions.

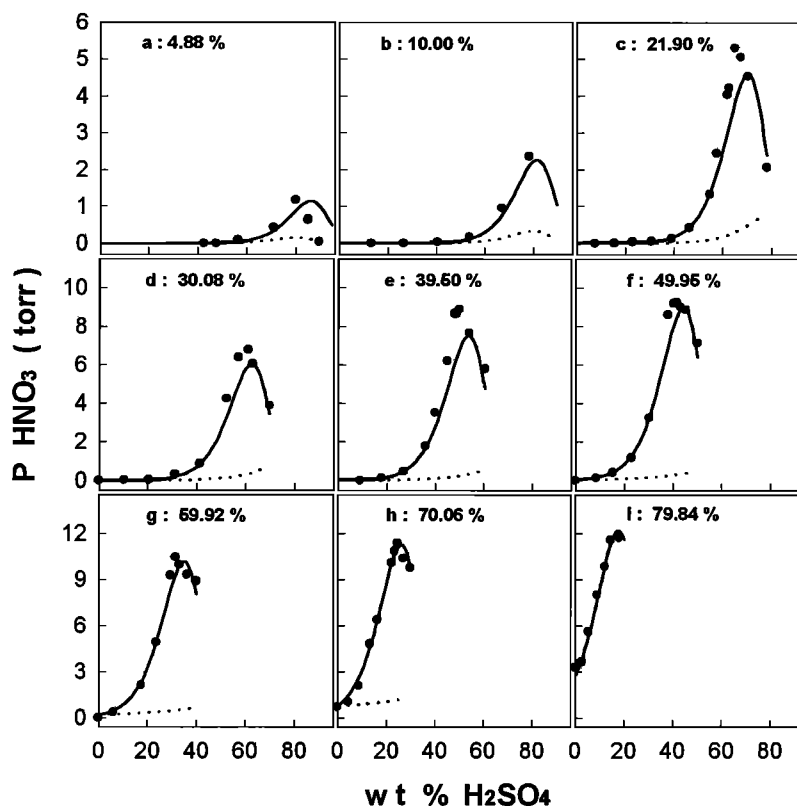


Figure 7. Comparison of the measured and predicted vapor pressures of HNO₃ over H₂O/HNO₃/H₂SO₄ mixtures at 273 K, plotted against wt % H₂SO₄. Weight percentages of HNO₃ are labeled in the figure. The symbols are Vandoni's [1944] experimental data, solid lines are our fitted values and dotted lines are Carslaw *et al.* [1995] predictions.

doing this, they ignored about 2/3 of the experimental determinations of Vandoni. Obviously, their model fails to represent higher vapor pressures. In addition, the partial pressures they predict for the ternary mixture do not tend smoothly to the values of the corresponding binary systems (see curves c-h of Figure 7).

The comparison at lower temperatures (Figures 8 and 9) is performed with the experimental results of Zhang *et al.* [1993b] and the parameterization of these results by Beyer *et al.* [1994]. Zhang *et al.* have determined H₂O and HNO₃ vapor pressures over HNO₃-H₂SO₄-H₂O mixtures, for temperatures ranging from less than 200 K to 240 K and for H₂SO₄ concentrations in the range 35-75 wt % and HNO₃ from 0.1 to 10 wt %. These data were later parameterized in order to expand their range of validity, by forcing them to be in agreement with the data of the binary HNO₃/H₂O system, in the limit that the sulfuric acid concentration is set to zero [Beyer *et al.*, 1994].

Our model results are shown as solid lines in Figures 8 and 9. The dashed lines are the experimental results from Zhang *et al.* (obtained from their Table I), and the dotted lines are the least squares fits from Beyer *et al.* (to avoid confusion, the experimental results are not always plotted, especially in the case of $P_{\text{H}_2\text{O}}$). As can be seen from these figures, the agreement between the model results and the parameterized curves is generally very good. For $P_{\text{H}_2\text{O}}$ the maximum deviation between the Beyer *et al.* parameterization and our model is 30%, over the full range of concentration of H₂SO₄ and HNO₃ and for the temperature range 190 to 273 K. This maximum deviation occurs

for 50 wt % of H₂SO₄ and 10 wt % of HNO₃ for temperatures around 273 K, while the agreement is much better at lower temperatures. This is, of course, due to the fact that the data of Zhang *et al.* have been collected for temperatures in the range 190-250 K. Since our model fits very well Vandoni's data at 273 K, one may speculate that the slope determined by Zhang *et al.* might be in (slight) error. For most of the other concentrations, the difference is better than 10%, i.e., of the order of the experimental uncertainties, and become generally better for lower temperatures.

For P_{HNO_3} , the maximum deviation, i.e., 50%, occurs for 35 wt % of H₂SO₄ and 5 wt % of HNO₃ and for 40 wt % of H₂SO₄ and 1 wt % of HNO₃. For most of the other concentrations, the differences remain below 25 %. Generally, the differences tend to be larger at low temperatures and seem to be due to an "excess of curvature" of the plots. This curvature itself seems to be connected with the fact that the coefficients $A_{\text{H}_2\text{O}}$ and A_{HNO_3} of the binary mixture HNO₃/H₂SO₄ have been considered as independent of the temperature, owing to the lack of experimental data.

The comparison with other models at low temperatures can be made with the work of Luo *et al.* [1995] and Tabazadeh *et al.* [1994]. Luo *et al.* developed an activity coefficient model, able to predict P_{HNO_3} and $P_{\text{H}_2\text{O}}$ (their equations 4 and 7). In their article, they compare only P_{HNO_3} with the data of Zhang *et al.* and found that, for H₂SO₄ concentration below 40%, the agreement is within 40%, but for higher concentrations, "model values up to factors of 2, 5, and 10 lower than the measurements in 50, 60, and 70 wt %

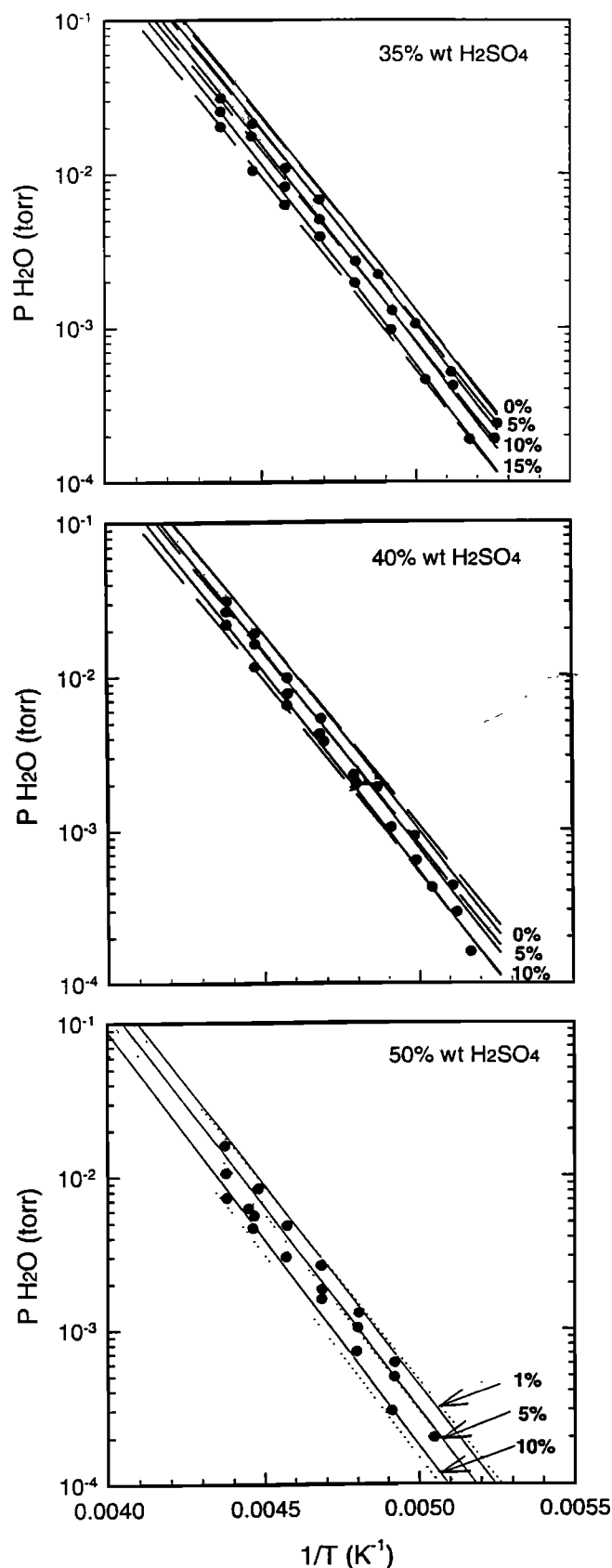


Figure 8. Water vapor pressures over H₂O/HNO₃/H₂SO₄ solutions. Weight percents of sulfuric acid and nitric acid are indicated in the figures. The solid lines are our predictions, symbols are Zhang *et al.* [1993b] measurements, and dotted lines are parameterized values of Beyer *et al.* [1994].

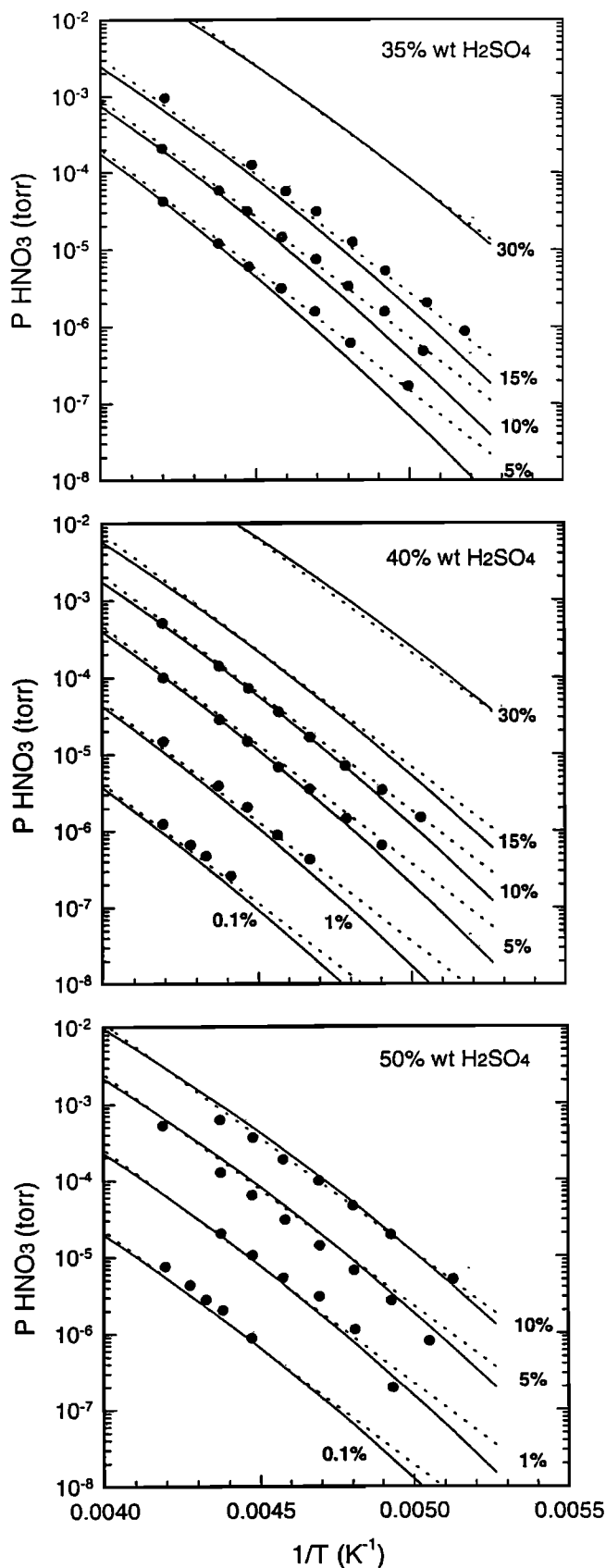


Figure 9. Nitric acid vapor pressures over H₂O/HNO₃/H₂SO₄ solutions. Weight percents of sulfuric acid and nitric acid are indicated in the figures. The solid lines are our predictions, symbols are Zhang *et al.* [1993b] measurements, and dotted lines are parameterized values of Beyer *et al.* [1994].

H₂SO₄ respectively." Recall that, even for 70 wt % H₂SO₄, our model shows only a 45% maximum difference with Zhang et al. data for P_{HNO_3} and a 20% maximum difference for $P_{\text{H}_2\text{O}}$.

Another model was also recently published by Tabazadeh et al. [1994]. Their model is based on the Kuzik and Meissner [1978] method, derived from Brønsted's theory of specific ionic interactions. Tabazadeh et al. compared in their Figure 4 their model predictions with Zhang et al. data for P_{HNO_3} . Although they do not quantify the differences, the agreement seems good for the curves depicted, i.e., for H₂SO₄ concentrations of 35, 40, and 50 wt % and for HNO₃ concentrations ranging from 1 to 15 wt %. We have in the past done similar calculations [Taleb, 1992] with the use of the Kuzik and Meissner method for the H₂O/HNO₃/H₂SO₄ system and found that, when compared with Vandoni's data at 273 K, the results are generally correct as long as the concentrations of H₂SO₄ and HNO₃ remain below 65 and 10%, respectively. This is due to the fact that this model neglect solute-solute interactions. Furthermore, we noted that this model is unable to predict the maximum observed for P_{HNO_3} and is also unable to predict directly the partial vapor pressure of sulfuric acid (of course, the use of Gibbs-Duhem equation makes possible the determination of $P_{\text{H}_2\text{SO}_4}$).

6. Conclusion

In this article, we have presented a simple thermodynamic model able to generate the equilibrium partial pressures over the binary H₂O/HNO₃, H₂O/H₂SO₄, HNO₃/H₂SO₄ mixtures and the ternary H₂O/HNO₃/H₂SO₄ system. These predictions are based on equations which appear to be much simpler and more precise than what has been published on this subject in the past. This is also the only model able to predict directly $P_{\text{H}_2\text{SO}_4}$. Although $P_{\text{H}_2\text{SO}_4}$ cannot be compared with any experimental determinations, we believe that the values are probably correct, since the equations used to generate them (equations (10a)-(10c)) are self-consistent, i.e., verify the Gibbs-Duhem equation. For the ternary system H₂O/HNO₃/H₂SO₄, the predictions concerning P_{HNO_3} can probably be improved if additional data for the binary HNO₃/H₂SO₄ mixture, such as the temperature dependence of P_{HNO_3} , are made available.

Appendix: Model Description

The model used in this article is developed from the article published by Li and Coull [1948] but is, in fact, based on the model of regular solutions developed, among others, by Van Laar [1910], Scatchard [1931], and Hildebrand and Scott [1936]. This method allows one to calculate the activities or activity coefficients in a ternary system, from the activities of the corresponding binary mixtures, provided that they can be described by a Van Laar type of equation (1a)-(1b). As we already noted [Jaecker-Voirol et al., 1990], these equations apply to mixtures of relatively simple, nonpolar liquids. However, it has been found that these equations are frequently able to represent activity coefficients of more complex solutions, if one regards the coefficients A and B as adjustable parameters.

The net change in internal energy due to the mixing of components 1 and 2 to form a binary solution is [Li and Coull, 1948]

$$\Delta E_{12} = \frac{a_{12}n_1n_2}{n_1V_1 + n_2V_2} \quad (11)$$

where n_1 , n_2 , V_1 , and V_2 are the number of moles per unit volume and the molar volumes of 1 and 2, respectively, and where a_{12} is a constant, at constant temperature.

In the case where the volume changes on mixing, the molar volumes can be replaced by the corresponding partial molar volumes $\bar{V}_1$ and $\bar{V}_2$

The Gibbs free energy change in mixing ΔG_{12} is calculated by using the following assumptions: (1) the volume change on mixing $\Delta(PV)$ is negligible in comparison with ΔE . Thus, $\Delta H = \Delta E$, (2) the entropy change on mixing is the same as for ideal solution, i.e.,

$$\Delta S_{12} = -R(n_1 \ln X_1 + n_2 \ln X_2) \quad (12)$$

where R is the gas constant. The Gibbs free energy change on mixing ΔG_{12} is then

$$\Delta G_{12} = \Delta H_{12} - T\Delta S_{12} = \Delta E_{12} + RT(n_1 \ln X_1 + n_2 \ln X_2) \quad (13)$$

and therefore the total Gibbs free energy of the mixture G_{12} is

$$G_{12} = n_1 G_1^0 + n_2 G_2^0 + \Delta E_{12} + RT(n_1 \ln X_1 + n_2 \ln X_2) \quad (14)$$

where G_1^0 and G_2^0 are the free energy of the pure components 1 and 2, respectively.

By definition of the partial molar Gibbs free energy $\bar{G}$ or chemical potential μ :

$$\mu_1 = \bar{G}_1 = \left(\frac{\partial G_{12}}{\partial n_1} \right)_{T,P,n_2}$$

one obtains

$$\bar{G}_1 = G_1^0 + RT \ln X_1 + \frac{\partial(\Delta E_{12})}{\partial n_1}$$

or

$$\mu_1 = \mu_1^0 + RT \ln X_1 + \frac{\partial(\Delta E_{12})}{\partial n_1} \quad (15)$$

Since we know that the chemical potential is related to the activity a_1 or the activity coefficient γ_1 through $\bar{G}_1 - G_1^0 = RT \ln a_1$, and $a_1 = \gamma_1 X_1$, we can deduce

$$RT \ln \gamma_1 = \frac{\partial}{\partial n_1}(\Delta E_{12}) \quad (16)$$

Note that, in the above formula, γ_1 is the activity coefficient in the mole fraction scale. Applying this last formula to Eq. (11), and defining $k_{12} = a_{12}/2.3R$, one obtains

$$T \log \gamma_1 = \frac{1}{\left[\left(\frac{\bar{V}_1}{\bar{V}_2} \right) \sqrt{\frac{\bar{V}_2}{k_{12}}} \left(\frac{X_1}{X_2} \right) + \sqrt{\frac{\bar{V}_2}{k_{12}}} \right]^2} \quad (17a)$$

Similarly,

$$T \log \gamma_2 = \frac{1}{\left[\left(\frac{\bar{V}_2}{\bar{V}_1} \right) \sqrt{\frac{\bar{V}_1}{k_{12}}} \left(\frac{X_2}{X_1} \right) + \sqrt{\frac{\bar{V}_1}{k_{12}}} \right]^2} \quad (17b)$$

Both equations (17a) and (17b) can easily be rearranged to give the familiar two-suffix Van Laar's equations (1a) and (1b) by

setting $A_1 = k_{12}/\bar{V}_2$, $A_2 = k_{12}/\bar{V}_1$, $B_1 = \bar{V}_1/\bar{V}_2$, and $B_2 = \bar{V}_2/\bar{V}_1$. Thus a plot of $(T \log \gamma_1)^{-1/2}$ versus X_1/X_2 gives a straight line having a slope $(\bar{V}_1/\bar{V}_2) (k_{12}/k_{23})^{1/2}$ and an intercept $(\bar{V}_2/k_{12})^{1/2}$. Once the slopes and intercepts are found, the constants A_1 , A_2 , B_1 , and B_2 are easily determined.

Equations (11)–(17) can easily be extended to ternary systems. In such a case, the internal energy change on mixing is

$$\Delta E_{123} = \frac{a_{12}n_1n_2 + a_{13}n_1n_3 + a_{23}n_2n_3}{n_1\bar{V}_1 + n_2\bar{V}_2 + n_3\bar{V}_3} \quad (18)$$

On substitution of equation (18) in equation (16), one obtains

$$RT \ln \gamma_1 = \frac{\partial}{\partial n_1} \left(\frac{a_{12}n_1n_2 + a_{13}n_1n_3 + a_{23}n_2n_3}{n_1\bar{V}_1 + n_2\bar{V}_2 + n_3\bar{V}_3} \right) \quad (19)$$

Again, setting $k_{ij} = a_{ij}/2.3R$, and after derivation, equation (19) becomes

$$\begin{aligned} T \log \gamma_1 &= \frac{k_{12}\bar{V}_2n_2^2 + k_{13}\bar{V}_3n_3^2 + (k_{12}\bar{V}_3 + k_{13}\bar{V}_2 - k_{23}\bar{V}_1)n_2n_3}{[\bar{V}_1n_1 + \bar{V}_2n_2 + \bar{V}_3n_3]^2} \\ &= \frac{k_{12}[\bar{V}_2n_2^2 + \bar{V}_3n_2n_3] - 0.5k_{23}\bar{V}_1n_2n_3}{[\bar{V}_1n_1 + \bar{V}_2n_2 + \bar{V}_3n_3]^2} \\ &\quad + \frac{k_{13}[\bar{V}_3n_3^2 + \bar{V}_2n_2n_3] - 0.5k_{23}\bar{V}_1n_2n_3}{[\bar{V}_1n_1 + \bar{V}_2n_2 + \bar{V}_3n_3]^2} \end{aligned} \quad (20)$$

Neglecting the effect of composition on partial molar volumes and dividing by the total number of moles, these equations can be rearranged as

$$\begin{aligned} T \log \gamma_1 &= \frac{\left(\frac{k_{12}}{\bar{V}_2}\right) \left[X_2^2 + \left(\frac{\bar{V}_3}{\bar{V}_2}\right) X_2X_3\right] - 0.5 \left(\frac{k_{23}}{\bar{V}_2}\right) \left(\frac{\bar{V}_1}{\bar{V}_2}\right) X_2X_3}{\left[\left(\frac{\bar{V}_1}{\bar{V}_2}\right) X_1 + X_2 + \left(\frac{\bar{V}_3}{\bar{V}_2}\right) X_3\right]^2} \\ &\quad + \frac{\left(\frac{k_{13}}{\bar{V}_3}\right) \left[X_3^2 + \left(\frac{\bar{V}_2}{\bar{V}_3}\right) X_2X_3\right] - 0.5 \left(\frac{k_{23}}{\bar{V}_3}\right) \left(\frac{\bar{V}_1}{\bar{V}_3}\right) X_2X_3}{\left[\left(\frac{\bar{V}_1}{\bar{V}_3}\right) X_1 + \left(\frac{\bar{V}_2}{\bar{V}_3}\right) X_2 + X_3\right]^2} \end{aligned} \quad (21a)$$

Similarly,

$$\begin{aligned} T \log \gamma_2 &= \frac{\left(\frac{k_{12}}{\bar{V}_1}\right) \left[X_1^2 + \left(\frac{\bar{V}_3}{\bar{V}_1}\right) X_1X_3\right] - 0.5 \left(\frac{k_{13}}{\bar{V}_1}\right) \left(\frac{\bar{V}_2}{\bar{V}_1}\right) X_1X_3}{\left[X_1 + \left(\frac{\bar{V}_2}{\bar{V}_1}\right) X_2 + \left(\frac{\bar{V}_3}{\bar{V}_1}\right) X_3\right]^2} \\ &\quad + \frac{\left(\frac{k_{23}}{\bar{V}_3}\right) \left[X_3^2 + \left(\frac{\bar{V}_1}{\bar{V}_3}\right) X_1X_3\right] - 0.5 \left(\frac{k_{13}}{\bar{V}_3}\right) \left(\frac{\bar{V}_2}{\bar{V}_3}\right) X_1X_3}{\left[\left(\frac{\bar{V}_1}{\bar{V}_3}\right) X_1 + \left(\frac{\bar{V}_2}{\bar{V}_3}\right) X_2 + X_3\right]^2} \end{aligned} \quad (21b)$$

$$\begin{aligned} T \log \gamma_3 &= \frac{\left(\frac{k_{13}}{\bar{V}_1}\right) \left[X_1^2 + \left(\frac{\bar{V}_2}{\bar{V}_1}\right) X_1X_2\right] - 0.5 \left(\frac{k_{12}}{\bar{V}_1}\right) \left(\frac{\bar{V}_3}{\bar{V}_1}\right) X_1X_2}{\left[X_1 + \left(\frac{\bar{V}_2}{\bar{V}_1}\right) X_2 + \left(\frac{\bar{V}_3}{\bar{V}_1}\right) X_3\right]^2} \\ &\quad + \frac{\left(\frac{k_{23}}{\bar{V}_2}\right) \left[X_2^2 + \left(\frac{\bar{V}_1}{\bar{V}_2}\right) X_1X_2\right] - 0.5 \left(\frac{k_{12}}{\bar{V}_2}\right) \left(\frac{\bar{V}_3}{\bar{V}_2}\right) X_1X_2}{\left[\left(\frac{\bar{V}_1}{\bar{V}_2}\right) X_1 + X_2 + \left(\frac{\bar{V}_3}{\bar{V}_2}\right) X_3\right]^2} \end{aligned} \quad (21c)$$

Equations (21a)–(21c), although based on the same principles as those developed by Li and Coull [1948], are somewhat different in form. Their main advantage is that, when one of the components is vanishing, they transform easily to the binary cases. Rewriting them in terms of our constants A and B , one obtains Equations (10a)–(10c).

Acknowledgment. This work was supported by the EU under contract EV 5V-CT91-0044.

References

- Ayers, G.P., R.W. Gillett, and J.L. Gras, On the vapor pressure of sulfuric acid, *Geophys. Res. Lett.*, **7**, 433–436, 1980.
- Beyer, K.D., S.W. Seago, H.Y. Chang, and M.J. Molina, Composition and freezing of aqueous H₂SO₄/HNO₃ solutions under polar stratospheric conditions, *Geophys. Res. Lett.*, **21**, 871–874, 1994.
- Bolsaitis, P., and J. F. Elliott, Thermodynamic activities and equilibrium partial pressures for the aqueous sulfuric acid solutions, *J. Chem. Eng. Data*, **35**, 69–85, 1990.
- Brimblecombe, P., and S.L. Clegg, Equilibrium partial pressures of strong acids over concentrated solutions, III, The temperature variation of HNO₃ solubility, *Atmos. Environ.*, **24A**, 1945–1955, 1990.
- Carlsaw, K.S., S.L. Clegg, and P. Brimblecombe, A thermodynamic model of the system HCl–HNO₃–H₂SO₄–H₂O, including solubilities of HBr, from <200 to 328 K, *J. Phys. Chem.*, **99**, 11,557–11,574, 1995.
- Clavelin, J.L., and P. Mirabel, Détermination des pressions partielles du mélange eau-acide nitrique, *J. Chim. Phys.*, **76**, 533–537, 1979.
- Clegg, S.L., and P. Brimblecombe, Equilibrium partial pressures and mean activity and osmotic coefficients of 0–100% nitric acid as a function of temperature, *J. Phys. Chem.*, **94**, 5369–5380, 1990.
- Clegg, S.L., and P. Brimblecombe, Application of a multicomponent thermodynamic model to activities and thermal properties of 0–40 mol kg^{−1} aqueous sulfuric acid from 200 to 328 K, *J. Chem. Eng. Data*, **40**, 43–64, 1995.
- Crutzen, P., and F. Arnold, Nitric acid cloud formation in the cold Antarctic stratosphere: A major cause for the springtime "ozone hole," *Nature*, **324**, 651–655, 1986.
- Duisman, J.A., and S.A. Stern, Vapor pressure and boiling point of pure nitric acid, *J. Chem. Eng. Data*, **14**, 457–459, 1969.
- Giauque, W. F., E.W. Hornung, J.E. Kunzler and T.R. Robin, The thermodynamic properties of aqueous sulfuric acid solutions and hydrates from 15 to 300 K, *J. Am. Chem. Soc.*, **82**, 62–70, 1960.
- Gmitro, J. L., and T. Vermeulen, Vapor liquid equilibria for aqueous sulfuric acid, *AIChE J.*, **10**, 740–746, 1964.
- Hanson, D., and K. Mauersberger, Vapor pressures of HNO₃/H₂O solutions at low temperatures, *J. Phys. Chem.*, **92**, 6167–6170, 1988.
- Hildebrand, J.H., and R.L. Scott, *The Solubility of Nonelectrolytes*, Dover, New York, N. Y., 1964.
- Jaeger-Voirol, A., J.L. Ponche, and P. Mirabel, Vapor pressures in the ternary system water–nitric acid–sulfuric acid at low temperatures, *J. Geophys. Res.*, **95**(D8), 11,857–11,863, 1990. (Correction, *J. Geophys. Res.*, **95**(D13), 22,565, 1990.)
- Kuzik, C.L., and H.P. Meissner, Electrolyte activity coefficients in inorganic processing, *AIChE Symp. Ser.*, **74**, 14–20, 1978.
- Li, Y.M., and J. Coull, The prediction of vapour–liquid equilibria for ternary mixtures from the data of binary systems, *J. Inst. Petrol.*, **34**, 692–704, 1948.

- Luo, B., K.S. Carslaw, T. Peter, and S.L. Clegg, Vapor pressures of H₂SO₄/HNO₃/HCl/HBr/H₂O solutions to low stratospheric temperatures, *Geophys. Res. Lett.*, **22**, 247-250, 1995.
- Pitzer, K.S., and J.M. Simonson, Thermodynamics of multicomponent, miscible, ionic systems: Theory and equations, *J. Phys. Chem.*, **90**, 3005-3009, 1986.
- Prausnitz, J.M., *Molecular Thermodynamics of Fluid-Phase Equilibria*, 224 pp., Prentice-Hall, Englewood Cliffs, N.J., 1969.
- Redlich, O., W.E. Gargrave, and W.D. Krostek, Thermodynamics of solutions, Consistency of sparse data, activities of nitric and perchloric acids, *Ind. Eng. Chem.*, **7**, 211-214, 1968.
- Sapozhnikov, A., Über die Eigenschaften der Salpeterschwefelsäuregemische, *Z. Phys. Chem.*, **49**, 697-708, 1904.
- Scatchard, G., Equilibria in nonelectrolyte solutions in relation to the vapor pressures and densities of the components, *Chem. Rev.*, **8**, 321-333, 1931.
- Tabata, S., A simple but accurate formula for the saturation vapor pressure over liquid water, *J. Appl. Meteorol.*, **12**, 1410-1412, 1973.
- Tabazadeh, A., R.P. Turco, and M.Z. Jacobsen, A model for studying the composition and chemical effects of stratospheric aerosols, *J. Geophys. Res.*, **99**(D6), 12,897-12,914, 1994.
- Taleb, D.E., Prédiction des équilibres liquide-vapeur d'un système ternaire à partir des systèmes binaires associés, Diplôme d'Etudes Approfondies, Univ. Louis Pasteur, Strasbourg, 1992.
- Toon, O.B., P. Hamill, R. Turco, and J. Pinto, Condensation of HNO₃ and HCl in the winter polar stratospheres, *Geophys. Res. Lett.*, **13**, 1284-1287, 1986.
- Vandoni, R., Contribution à l'étude des mélanges sulfonitriques, *Mem. Serv. Chim. Etat*, **31**, 87-111, 1944.
- Vandoni, R., and M. Laudy, Mesure des tensions de vapeur partielles des mélanges NO₃H-H₂O et vérification de l'équation de Margules-Duhem, *J. Chim. Phys.*, **49**, 99-105, 1952.
- Van Laar, J.J., Über Dampfspannungen von Binären Gemischen, *Z. Phys. Chem.*, **72**, 723-751, 1910.
- Zelevnik, F.J., Thermodynamic properties of the aqueous sulfuric acid system to 350 K, *J. Phys. Chem. Ref. Data*, **20**, 1157-1200, 1991.
- Zhang, R., P.J. Wooldridge, J. P. D. Abbatt, and M.J. Molina, Physical chemistry of the binary system at low temperature: Stratospheric implications, *J. Phys. Chem.*, **97**, 7351-7358, 1993a.
- Zhang, R., P.J. Wooldridge, and M.J. Molina, Vapor pressure measurements for the H₂SO₄/HNO₃/H₂O and H₂SO₄/HCl/H₂O systems: Incorporation of stratospheric acids into background sulfate aerosols, *J. Phys. Chem.*, **97**, 8541-8548, 1993b.

P. Mirabel, J.-L. Ponche, and D.-E. Taleb, Centre de Géochimie de la Surface and Chemistry Department, Université Louis Pasteur, 28, rue Goethe, Strasbourg 67000, France. (e-mail:mirabel@illite.u-strasbg.fr)

(Received January 9, 1996; revised July 11, 1996; accepted July 11, 1996.)